

1 Model identity

FruitCropXL

A modular functional-structural fruit-crop model implemented in GroIMP / XL

Links 3D architecture, environment and management to fruit quality, organ growth and coupled water-carbon dynamics through stomatal regulation



2 Biological structure



- Represents interconnected 3D plant structure
- Topology and geometry drive local microclimate and resource dynamics

3 Inputs and configuration

model.options*.json
species, module switches, simulation time, outputs and management controls

plant.parameters*.json
phenology, growth, water and carbon related parameters

initial.conditions*.json
initial values for fruit/root/wood/leaf

soil-parameters*.json
soil layers, physical and chemical properties



Weather —
radiation, temperature, humidity, CO₂, soil water



Orchard design —
spacing, row orientation, training system



Plant architecture —
architecture file, root distribution

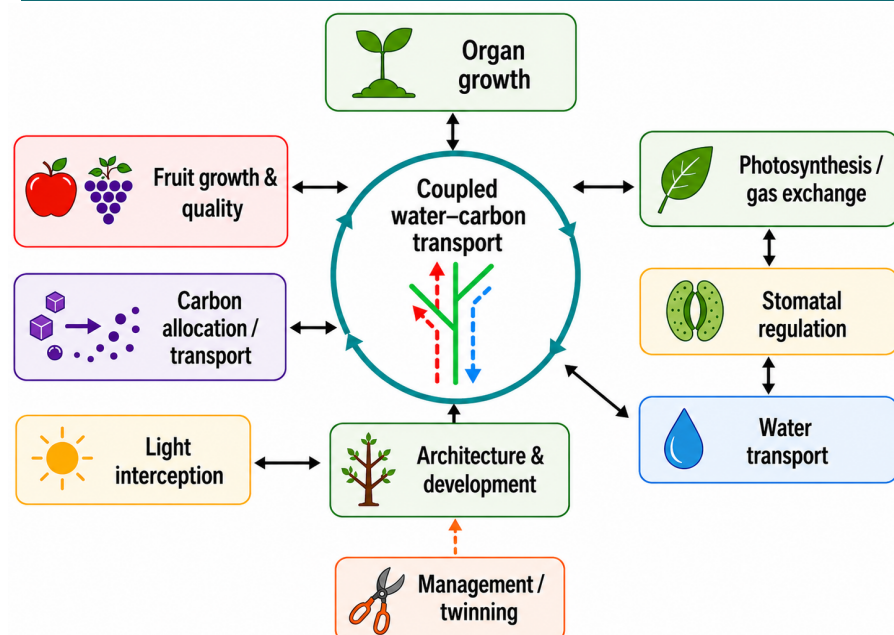


Management —
pruning, twinning

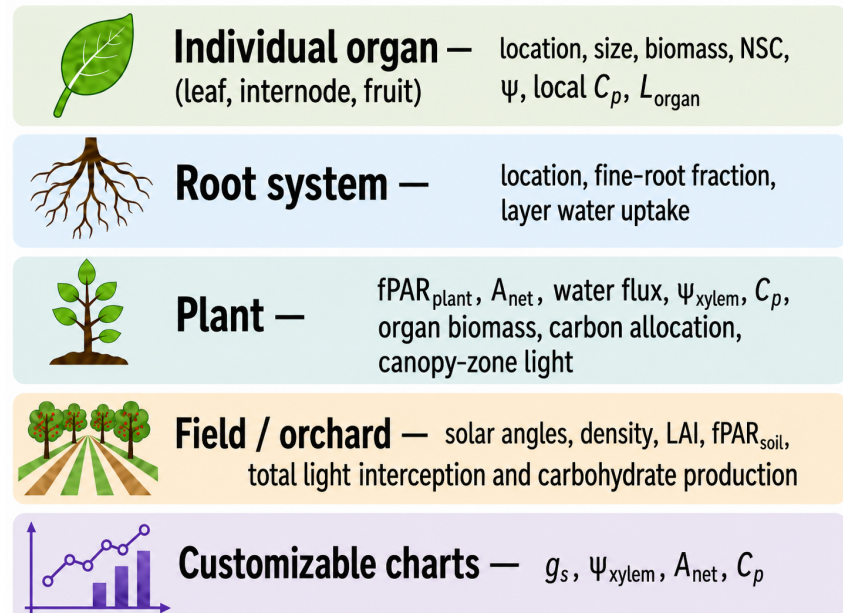


Support data —
organ-size profiles, shading factors

4 Core processes and modules



5 Output Layers



6 Intended use



Orchard layout/architecture/environment → fPAR, A_{net} , water flux, ψ and C_p distribution



Water × carbon status → fruit growth and quality



Virtual experiments across climate, water, crop load and cultivars



Explore ideotype and system design



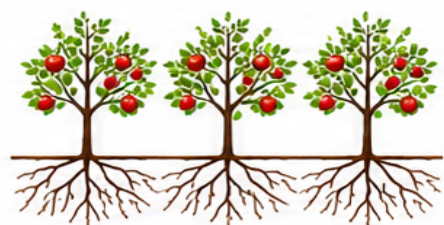
Digital-twin workflows



Modular extension, hypothesis testing, and surrogate model training



Configuration + inputs



3D plant/root structure



coupled processes



multi-scale outputs



use cases